SOLUTIONS OF A LOCALLY GIVEN SYSTEM ARE UNIQUE UP TO UNIQUE ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Map from a subcollection of opens. Topology, Lemma topology-lemma-create-map-from-subcollection. Continuous f : X -> Y with the given opens as inverse images of a basis.
- Sheaf rule for Hom presheaves. Cohomology of Sheaves, Lemma cohomology-lemma-vanishing-and-glueing. V |-> Hom(K|_{f^{-1}V}, K'|_{f^{-1}V}) is a sheaf.

Source reference. Original reference 0D67: sources/stacks-chow/tags.txt, source lines 16512–16512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D5Q: sources/stacks-chow/tags.txt, source lines 16495–16495. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D66: sources/stacks-chow/tags.txt, source lines 16511–16511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0D67 assume*

(1) $X = \bigcup_{U \in \mathcal{B}} U$ *and for* $U, V \in \mathcal{B}$ *we have* $U \cap V = \bigcup_{W \in \mathcal{B}, W \subset U \cap V} W$,
(2) *for any* $U \in \mathcal{B}$ *we have* $\mathrm{Ext}^i(K_U, K_U) = 0$ *for* $i < 0$.

If a solution (K, ρ_U) *exists, then it is unique up to unique isomorphism and moreover* $\mathrm{Ext}^i(K, K) = 0$ *for* $i < 0$.

Proof. Let (K, ρ_U) and (K', ρ'_U) be a pair of solutions. Let $f : X \to Y$ be the continuous map constructed in Topology, Lemma 0D5Q. Set $\mathcal{O}_Y = f_*\mathcal{O}_X$. Then K, K' and $\mathcal{B}$ are as in Lemma 0D66 part (2). Hence we obtain the vanishing of negative exts for K and we see that the rule

$$V \longmapsto \mathrm{Hom}(K|_{f^{-1}V}, K'|_{f^{-1}V})$$

is a sheaf on Y. As both (K, ρ_U) and (K', ρ'_U) are solutions the maps

$$(\rho'_U)^{-1} \circ \rho_U : K|_U \longrightarrow K'|_U$$

over $U = f^{-1}(f(U))$ agree on overlaps. Hence we get a unique global section of the sheaf above which defines the desired isomorphism $K \to K'$ compatible with all structure available. □

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